

Asymptotic behaviour of the Bardeen–Cooper–Schrieffer theory of superconductivity

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Superconductivity in experiments

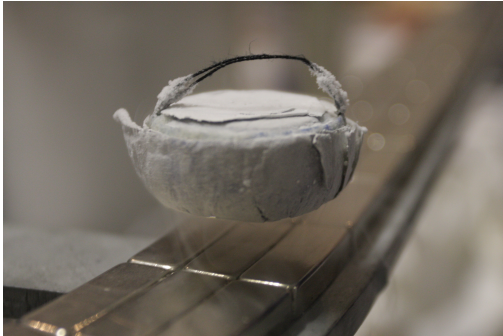


Figure: Levitating type 2 superconductor above permanent magnets. [Wikimedia commons]

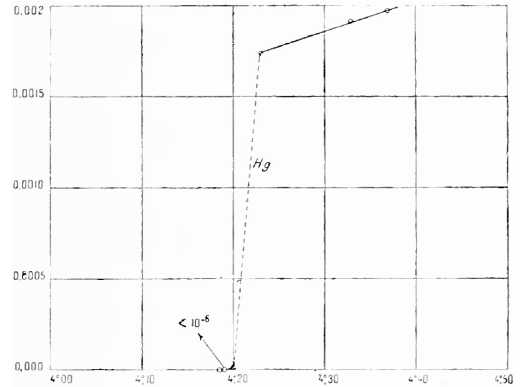


Figure: Resistivity vs. temperature of mercury.
First experimental observation of superconductivity.
[Kammerlingh Onnes 1911]

Bardeen–Cooper–Schrieffer (BCS) 1957 theory of superconductivity

Described by (multi-band) **BCS gap equation**

$$\Delta_a(p) = -\frac{1}{(2\pi)^{d/2}} \sum_{b=1}^n \int_{\mathbb{R}^d} \hat{V}_{ab}(p-q) \frac{\Delta_b(q)}{K_{T,b}^{\Delta_b}} dq, \quad a = 1, \dots, n, \quad K_{T,a}^{\Delta_a}(p) = \frac{\sqrt{\varepsilon_a(p)^2 + |\Delta_a(p)|^2}}{\tanh\left(\frac{\sqrt{\varepsilon_a(p)^2 + |\Delta_a(p)|^2}}{2T}\right)}.$$

n : Number of bands (for most of the talk $n = 1$)

V_{ab} : Effective interactions. **Possibly attractive**, meaning V has negative part,

ε_a : Effective dispersion relations, e.g. Sommerfeld: $\varepsilon_a(p) = p^2/2m_a - \mu_a$.

Material is superconducting if non-zero solution $\Delta = (\Delta_a)_{a=1}^n$ exists.

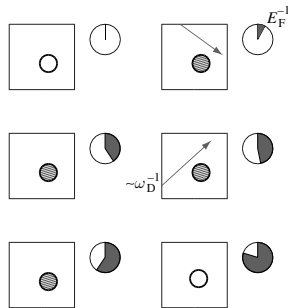
In matrix-valued form:

$$\Delta = -\frac{1}{(2\pi)^{d/2}} \hat{V} * (K_T^{\Delta})^{-1} \Delta, \quad K_T^{\Delta} = \text{diag}[K_{T,a}^{\Delta_a}].$$

Arises as Euler–Lagrange equation of **BCS functional** (with

$$\Delta_a = -2 \sum_b \widehat{V_{ab} \alpha_b})$$

$$\mathcal{F}(\gamma, \alpha) = \int_{\mathbb{R}^d} \sum_{a=1}^n \varepsilon_a(p) \gamma_a(p) dp + \int_{\mathbb{R}^d} \sum_{a,b=1}^n V_{ab}(x) \overline{\alpha_a(x)} \alpha_b(x) dx - TS(\gamma, \alpha).$$



Heuristic derivation of the BCS functional (one-band case)

Consider interacting many-body fermionic system on some box $[0, L]^3$ with periodic boundary conditions

$$H_N = \sum_{j=1}^N \varepsilon(-i\nabla_{x_j}) + \sum_{1 \leq j < k \leq N} V(x_j - x_k) \quad \rightsquigarrow \quad \mathcal{H} = \sum_{p, \sigma} \varepsilon(p) a_{p, \sigma}^* a_{p, \sigma} + \sum_{p, q, r} \sum_{\sigma, \tau} \hat{V}(r) a_{p+r, \sigma}^* a_{q-r, \tau}^* a_{q, \tau} a_{p, \sigma}$$

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Consider quasi-free states ρ . Wick rule:

$$\langle a_{p+r, \sigma}^* a_{q-r, \tau}^* a_{q, \tau} a_{p, \sigma} \rangle_\rho = \langle a_{p+r, \sigma}^* a_{p, \sigma} \rangle_\rho \langle a_{q-r, \tau}^* a_{q, \tau} \rangle_\rho - \langle a_{p+r, \sigma}^* a_{q, \tau} \rangle_\rho \langle a_{q-r, \tau}^* a_{p, \sigma} \rangle_\rho + \langle a_{p+r, \sigma}^* a_{q-r, \tau}^* \rangle_\rho \langle a_{q, \tau} a_{p, \sigma} \rangle_\rho.$$

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Drop first two terms and assume translation- and $SU(2)$ -invariance (spin singlet superconductivity).

$$\rightsquigarrow \text{only non-zero terms are} \quad \hat{\gamma}(p) = \langle a_{p, \uparrow}^* a_{p, \uparrow} \rangle_\rho = \langle a_{p, \downarrow}^* a_{p, \downarrow} \rangle_\rho, \quad \hat{\alpha}(p) = \langle a_{-p, \uparrow} a_{p, \downarrow} \rangle_\rho = -\langle a_{-p, \downarrow} a_{p, \uparrow} \rangle_\rho.$$

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Free energy is then (with formal thermodynamic limit) (ρ the state with 1-pdm (γ, α))

$$\mathcal{F}(\gamma, \alpha) = \frac{1}{L^3} \langle \mathcal{H} + T \log \rho \rangle_\rho = \int \varepsilon(p) \hat{\gamma}(p) dp + \iint \overline{\hat{\alpha}(p)} \hat{V}(p - q) \hat{\alpha}(q) dp dq - TS(\gamma, \alpha).$$

In multi-band case add appropriate band-indices.

Problem (Challenge assumptions)

Translation invariance [Deuchert–Geisinger–Hainzl–Loss 2018; Roos 2025]

Dropping direct and exchange terms [Bräunlich–Hainzl–Seiringer 2014]

Theorem (Hainzl–Hamza–Seiringer–Solovej 2008)

Let $V_{ab} = V_{ba} \in L^{p_d}$ be real-valued and even ($p_3 = 3/2$, $p_2 > 1$, $p_1 = 1$), $\mu \in \mathbb{R}$ and $0 \leq T < \infty$. Then the following are equivalent

1. A minimizer of the BCS functional has $\alpha \neq 0$,
2. There exists $\alpha \neq 0$ with $(K_T^\Delta + V)\alpha = 0$, (equivalently $\Delta = -2\widehat{V\alpha}$ satisfies the BCS gap equation),
3. The (matrix-valued) operator $K_T^0 + V$ has at least one negative eigenvalue.

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$\leadsto$ Define the **critical temperature** $T_c := \inf\{T \geq 0 : K_T^0 + V \geq 0\}$.

Phase transition at T_c : superconducting for $T < T_c$ and normal metal for $T \geq T_c$. (Follows easily by K_T monotone increasing in T .)

As operators $K_T^0 \geq 2T$. $\leadsto$ Need V non-positive (as matrix-valued operator) for $T_c > 0$.

$E_\Delta(p) = \sqrt{\varepsilon(p)^2 + |\Delta(p)|^2}$ dispersion relation for associated quasi-particles. (One-band case)

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T_c and Ξ are material-dependent (ε, V -dependent). Both measure “stability” of superconducting phase.

Question

- Formula for T_c and Ξ in various limits?
- Relation between T_c and Ξ ?

Zero temperature

Physics prediction for T_c small:

$$\frac{\Xi(T=0)}{T_c} \approx \pi e^{-\gamma} \approx 1.76$$

Well-known in physics literature.

Valid for s -wave superconductivity (Δ radial),
(Ex: V radial with $\hat{V} \leq 0$)

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- ▶ Weak coupling λV with $\lambda \rightarrow 0$,
- ▶ Low density $\mu \rightarrow 0$,
- ▶ High density $\mu \rightarrow \infty$.

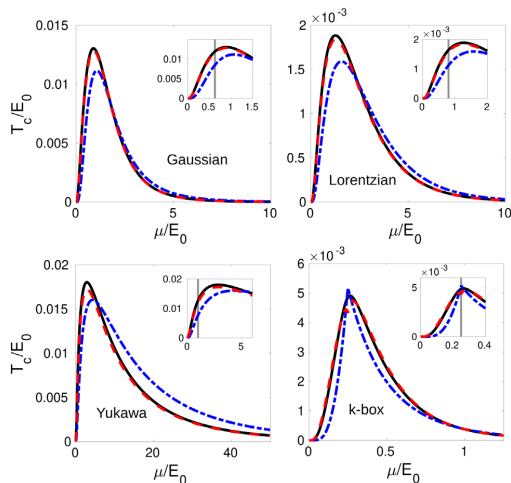


Figure: [Langmann–Triola–Balatsky 2019]

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Question

Can we find T_c and $\Xi(T=0)$ in these limits?

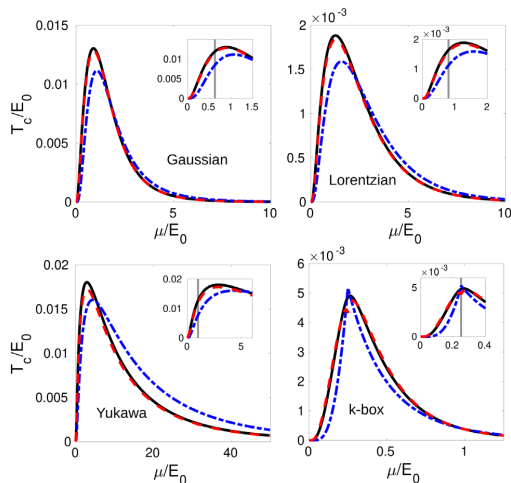


Figure: [Langmann–Triola–Balatsky 2019]

Theorem

Suppose V radial and $\hat{V} \leq 0$ (s -wave superconductivity). Then, T_c and $\Xi(T=0)$ satisfy

$$\frac{\Xi}{T_c} = \pi e^{-\gamma}(1 + o(1))$$

- ▶ in dimensions $d = 1, 2, 3$ for interaction λV in limit $\lambda \rightarrow 0$ (weak coupling) [Frank–Hainzl–Naboko–Seiringer 2007; Hainzl–Seiringer 2008b, Henheik–L.–Roos 2024],
- ▶ in dimension $d = 3$ in limit $\mu \rightarrow 0$ (low density) [Hainzl–Seiringer 2008a, L. 2021],
- ▶ in dimension $d = 3$ in limit $\mu \rightarrow \infty$ (high density) [Henheik 2022, Henheik–L. 2022],
- ▶ in dimension $d = 3$ in “any” limit $T_c, \Xi \rightarrow 0$ [Langmann–Triola–Balatsky 2019; Langmann–Triola 2023].

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Proofs by separate asymptotic formulas for T_c and Ξ in the various limits. E.g. low density ($\mu \rightarrow 0$) in dimension $d = 3$ ($a < 0$ is scattering length of $2V$)

$$T_c = \mu \left(\frac{8e^{\gamma-2}}{\pi} + o(1) \right) \exp \left(\frac{\pi}{2\sqrt{\mu}a} \right), \quad \Xi = \mu \left(8e^{-2} + o(1) \right) \exp \left(\frac{\pi}{2\sqrt{\mu}a} \right).$$

Non-zero temperature

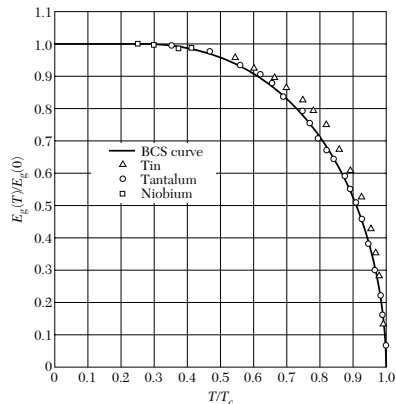
Physics prediction for T_c small (for s -wave superconductivity):

$$\frac{\Xi(T)}{T_c} \approx f_{\text{BCS}} \left(\sqrt{1 - \frac{T}{T_c}} \right)$$

for **universal** function f_{BCS} . For $T \rightarrow T_c$:

$$\frac{\Xi(T)}{T_c} \approx \sqrt{\frac{8\pi^2}{7\zeta(3)}} \sqrt{1 - \frac{T}{T_c}} \approx 3.06 \sqrt{1 - \frac{T}{T_c}}.$$

(In figure : $E_g(T) = \Xi(T)$)



Theorem (Langmann–Triola 2023, Henheik–L. 2025)

Picture holds

Non-zero temperature

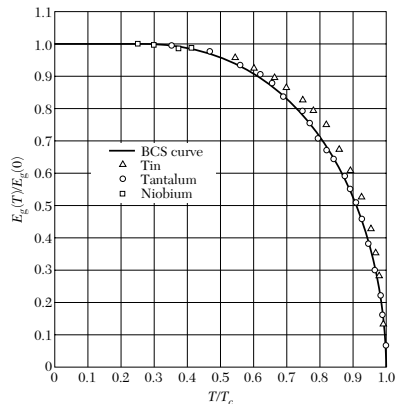
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Theorem (Langmann–Triola 2023, Henheik–L. 2025)

Picture holds for interaction λV radial in limit $\lambda \rightarrow 0$

- ▶ in dimensions $d = 1, 2, 3$ for V with $\hat{V} \leq 0$ (s -wave superconductivity, weak coupling limit),
- ▶ in dimension $d = 2$ for V with $\dim \ker(K_{T_c} + \lambda V) \leq 2$ (pure angular momentum, weak coupling limit),

Near the critical temperature T_c — Ginzburg–Landau theory

For $T \approx T_c$ BCS theory reduces to simpler Ginzburg–Landau (GL) theory (1950):

Theorem ([Frank–Hainzl–Seiringer–Solovej 2012; Frank–Lemm 2016])

For $T \rightarrow T_c$ we have

- ▶ $\min_{\gamma, \alpha} \mathcal{F}(\gamma, \alpha) = \min_{\gamma} \mathcal{F}(\gamma, 0) + (1 - T/T_c)^2 \min_{\psi \in \mathbb{C}^N} \mathcal{E}_{\text{GL}}(\psi) + O((T_c - T)^3)$ Here
 - ▶ $\mathcal{E}_{\text{GL}}(\psi)$ is a fourth order polynomial in $\psi, \bar{\psi} \in \mathbb{C}^N$, (for $N = 1$: $\mathcal{E}_{\text{GL}}(\psi) = C_1|\psi|^4 - C_2|\psi|^2$)
 - ▶ $N = \dim \ker K_{T_c} + V$,
- ▶ If (γ, α) minimize $\mathcal{F}$ then $\alpha \approx \sqrt{1 - T/T_c} |\psi_0| \mathfrak{a}$, with ψ_0 minimizer of $\mathcal{E}_{\text{GL}}$ and $\mathfrak{a} \in \ker K_{T_c} + V$ appropriate.

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Corollary

$$\lim_{\lambda \rightarrow 0} \lim_{T \rightarrow T_c} \frac{\Xi(T)/T_c}{\sqrt{1 - \frac{T}{T_c}}} = \sqrt{\frac{8\pi^2}{7\zeta(3)}}$$

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↪ Analysis of $\Xi(T)/T_c$ near T_c is “easy”. Away from T_c we need a different idea.

Multi-band effects

To probe **inter-band** effects we introduce coupling parameters ($V_{ab}^{\text{d}} = \delta_{ab} V_{aa}$, $V_{ab}^{\text{od}} = (1 - \delta_{ab}) V_{ab}$.)

$$V \longrightarrow \lambda V^{\text{d}} + \kappa \lambda V^{\text{od}}, \quad \lambda > 0, \quad \kappa \in \mathbb{R}.$$

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Inequality sharp for $\kappa \notin [-\kappa_c^-(\lambda), \kappa_c^+(\lambda)]$, for some $0 \leq \kappa_c^\pm(\lambda) < \infty$.*

In particular: Superconductivity for **purely repulsive** interactions for $|\kappa|$ large enough.

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Important role played by operators “ V on the Fermi surface(s)”

$$\mathcal{V}_{ab} = \mathfrak{F}_a V_{ab} \mathfrak{F}_b^\dagger : L^2(S_b) \rightarrow L^2(S_a), \quad \mathfrak{F}_a: \text{Fourier transform restricted to } S_a = \{p : \varepsilon_a(p) = 0\}, \quad \varepsilon_a = \inf \text{spec } \mathcal{V}_{aa}.$$

Theorem (Frank–Hainzl–Naboko–Seiringer 2007; Hainzl–Seiringer 2010)

Suppose $\min_{a \in \{1, \dots, n\}} \varepsilon_a < 0$. Then, $T_c(\lambda, 0) \sim \exp\left(\frac{1}{\lambda \min_a \varepsilon_a}\right)$ as $\lambda \rightarrow 0$

Theorem (Henheik–Langmann–L. 2024+)

Suppose $\min_{a \in \{1, \dots, n\}} \epsilon_a < 0$. Then, for $\lambda, |\kappa| \ll 1$:

(linear) If at least 2 minimizers $\hat{a}_1, \hat{a}_2 \in \{1, \dots, n\}$ of $a \mapsto \epsilon_a$:

$$T_c(\lambda, \kappa) = T_c(\lambda, 0) \exp \left[\frac{A_1^{\text{sgn}(\kappa)}}{\lambda^2} |\kappa \lambda| [1 + O(\lambda) + O(\kappa)] + O(e^{-c/\lambda}) \right], \quad A_1^\pm > 0.$$

If exactly 2 minimizers $\hat{a}_1, \hat{a}_2$ and $\dim \ker(\mathcal{V}_{\hat{a}_i \hat{a}_i} - \epsilon_{\hat{a}_i}) = 1$ for at least one $i = 1, 2$, then $A_1^+ = A_1^-$.
(Otherwise generically $A_1^+ \neq A_1^-$.)

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If exactly 2 minimizers $\hat{a}_1, \hat{a}_2$ and $\dim \ker(\mathcal{V}_{\hat{a}_i \hat{a}_i} - \epsilon_{\hat{a}_i}) = 1$ for at least one $i = 1, 2$, then $A_1^+ = A_1^-$.
(Otherwise generically $A_1^+ \neq A_1^-$.)

(quadratic) If minimizer $\hat{a}$ of $a \mapsto \epsilon_a$ is unique:

$$T_c(\lambda, \kappa) = T_c(\lambda, 0) \exp \left[\frac{A_2}{\lambda^3} (\kappa \lambda)^2 [1 + O(\lambda) + O(\kappa)] + O(e^{-c/\lambda}) \right], \quad A_2 > 0.$$

Theorem (Henheik–Langmann–L. 2024+)

Suppose $\min_{a \in \{1, \dots, n\}} \epsilon_a < 0$. Then, for $\lambda, |\kappa| \ll 1$:

(linear) If at least 2 minimizers $\hat{a}_1, \hat{a}_2 \in \{1, \dots, n\}$ of $a \mapsto \epsilon_a$:

$$T_c(\lambda, \kappa) = T_c(\lambda, 0) \exp \left[\frac{A_1^{\text{sgn}(\kappa)}}{\lambda^2} |\kappa \lambda| [1 + O(\lambda) + O(\kappa)] + O(e^{-c/\lambda}) \right], \quad A_1^\pm > 0.$$

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Recall $T_c \sim \exp(1/\lambda \epsilon_{\min}(\kappa))$, $\epsilon_{\min}(\kappa) = \inf \text{spec } \mathcal{V}(\kappa)$, $\mathcal{V}(\kappa) = \mathcal{V}^d + \kappa \mathcal{V}^{\text{od}}$.

Example (Idea behind ‘linear’ versus ‘quadratic’)

Consider eigenvalues of 2×2 matrices

$$\begin{bmatrix} 1 & \kappa \\ \kappa & 1 \end{bmatrix}, \quad \lambda_{\max} = 1 + |\kappa|, \quad \text{and} \quad \begin{bmatrix} 1 & \kappa \\ \kappa & 0 \end{bmatrix}, \quad \lambda_{\max} = \frac{1 + \sqrt{1 + 4\kappa^2}}{2} = 1 + \kappa^2 + O(\kappa^4).$$

Proposition (Henheik–Langmann–L. 2024+)

For any $\lambda > 0$ and $\kappa \in \mathbb{R}$ we have $T_c(\lambda, \kappa) \geq T_c(\lambda, 0)$. (Possibly $T_c(\lambda, 0) = 0$.)
Inequality sharp for $\kappa \notin [-\kappa_c^-(\lambda), \kappa_c^+(\lambda)]$, for some $0 \leq \kappa_c^\pm(\lambda) < \infty$.

Recall: $T_c(\lambda, \kappa) = \inf\{T \geq 0 : K_T + \lambda V^d + \lambda \kappa V^{\text{od}} \geq 0\}$.

Proof.

Note first $K_{T_c(\lambda, 0)} + \lambda V^d \geq 0$ by definition of T_c .

Define by variational principle

$$f(\kappa) = \inf \text{spec}(K_{T_c(\lambda, 0)} + \lambda V^d + \lambda \kappa V^{\text{od}}) = \inf_{\psi: \|\psi\|_{L^2} = 1} \langle \psi | K_{T_c(\lambda, 0)} + \lambda V^d + \lambda \kappa V^{\text{od}} | \psi \rangle.$$

f is a concave function of κ .

Since V^{od} is off-diagonal, f has maximum at $\kappa = 0$. (Choose ψ optimizer for $\kappa = 0$; ψ lives in only one band.)

By definition of T_c : $f(\kappa) < 0 \iff T_c(\lambda, \kappa) > T_c(\lambda, 0)$.

Thus, by concavity, $T_c(\lambda, \kappa) \geq T_c(\lambda, 0)$ with sharp inequality outside some interval of κ 's.

Interval is finite: Note that we can find ψ 's with $\langle \psi | K_{T_c(\lambda, 0)} + \lambda V^d | \psi \rangle \leq C$ and $\langle \psi | V^{\text{od}} | \psi \rangle \leq -c < 0$ or $\langle \psi | V^{\text{od}} | \psi \rangle \geq c > 0$. (By V^{od} off-diagonal; hence having negative spectrum.) □

Main ideas of proofs — Birman–Schwinger principle

“ $\Delta \neq 0$ solves BCS gap equation”

$\Leftrightarrow$ “ $K_T^\Delta + V$ has lowest eigenvalue 0 and associated eigenvector $\alpha = -\frac{\Delta}{2K_T^\Delta}$ ”

$\Leftrightarrow$ “Birman–Schwinger operator

$$B_{T,\Delta} = \operatorname{sgn} V |V|^{1/2} (K_T^\Delta)^{-1} |V|^{1/2}$$

has -1 as lowest eigenvalue and associated eigenvector $\phi = \operatorname{sgn} V |V|^{1/2} \alpha$.”

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To study $B_{T,\Delta}$ define (recall $K_T^\Delta = \frac{\sqrt{(p^2 - \mu)^2 + |\Delta(p)|^2}}{\tanh \frac{\sqrt{(p^2 - \mu)^2 + |\Delta(p)|^2}}{2T}}$, here assuming $\varepsilon(p) = p^2 - \mu$)

$$m(T, \Delta) = \frac{1}{|\mathbb{S}^{d-1}|} \int_{|p| \leq \sqrt{2\mu}} \frac{1}{K_T^\Delta(p)} dp$$

Decompose (with $\mathfrak{F}_\mu : L^1(\mathbb{R}^d) \rightarrow L^2(\mathbb{S}^{d-1})$ a rescaled Fourier transform)

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$B_{T,\Delta}$ has -1 as lowest eigenvalue $\leadsto$

$$m(T, \Delta) = - \left[\inf \operatorname{spec} \left(\mathfrak{F}_\mu |V|^{1/2} \frac{1}{1 + \operatorname{sgn} V |V|^{1/2} M_{T,\Delta} |V|^{1/2}} \operatorname{sgn} V |V|^{1/2} \mathfrak{F}_\mu^\dagger \right) \right]^{-1}.$$

$$m(T, \Delta) = - \left[\inf \text{spec} \left(\mathfrak{F}_\mu |V|^{1/2} \frac{1}{1 + \text{sgn } V |V|^{1/2} M_{T, \Delta} |V|^{1/2} \text{sgn } V |V|^{1/2} \mathfrak{F}_\mu^\dagger} \right) \right]^{-1}.$$

Evaluate directly $m(T, \Delta) = \frac{1}{|\mathbb{S}^{d-1}|} \int_{|p| \leq \sqrt{2\mu}} \frac{1}{\kappa_T^\Delta(p)} \, dp$ at $T = 0$ and $T = T_c$:

$$m(T = 0, \Delta) \approx \mu^{d/2-1} \left[\log \frac{\mu}{\Xi(T = 0)} + \log(2c_d) \right], \quad m(T_c, \Delta = 0) \approx \mu^{d/2-1} \left[\log \frac{\mu}{T_c} + \log \frac{e^\gamma}{\pi} + \log(2c_d) \right],$$

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Zero-temperature: Neglect error term $M_{T,\Delta}$ and consider for $T = 0$ and $T = T_c$:

$$m(T = 0, \Delta) \approx m(T_c, \Delta = 0) \approx \frac{-1}{\mathfrak{e}_\mu}, \quad \mathfrak{e}_\mu = \inf \text{spec} \left(\mathfrak{F}_\mu V \mathfrak{F}_\mu^\dagger \right) < 0.$$

Thus,

$$T_c \approx 2c_d \frac{e^\gamma}{\pi} \exp \left(\frac{1}{\mu^{d/2-1} \mathfrak{e}_\mu} \right), \quad \Xi(T = 0) \approx 2c_d \exp \left(\frac{1}{\mu^{d/2-1} \mathfrak{e}_\mu} \right),$$

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Zero-temperature: Neglect error term $M_{T,\Delta}$ and consider for $T = 0$ and $T = T_c$:

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Positive temperature: Neglect error term $M_{T,\Delta}$ and consider for T and $T = T_c$:

$$\int_0^{\sqrt{2\mu}} \left(\frac{1}{K_T^\Delta(p)} - \frac{1}{K_{T_c}^0(p)} \right) |p|^{d-1} dp = m(T, \Delta) - m(T_c, \Delta = 0) \approx 0$$

Essentially defining equation for $f_{\text{BCS}} \rightsquigarrow \Xi(T)/T_c \approx f_{\text{BCS}}(\sqrt{1 - T/T_c})$.

$$m(T, \Delta) = - \left[\inf \text{spec} \left(\mathfrak{F}_\mu |V|^{1/2} \frac{1}{1 + \text{sgn } V |V|^{1/2} M_{T, \Delta} |V|^{1/2}} \text{sgn } V |V|^{1/2} \mathfrak{F}_\mu^\dagger \right) \right]^{-1},$$

$$m(T_c, \Delta = 0) \approx \mu^{d/2-1} \left[\log \frac{\mu}{T_c} + \log \frac{e^\gamma}{\pi} + \log(2c_d) \right].$$

Multi-band: Neglect error term $M_{T, \Delta}$ and consider for $\kappa = 0$ and $\kappa \neq 0$:

$$\begin{aligned} \mu^{d/2-1} \log \frac{T_c(\kappa)}{T_c(0)} &\approx m(T_c(\kappa = 0), \Delta = 0) - m(T_c(\kappa), \Delta = 0) \\ &\approx - \frac{1}{\inf \text{spec}(\mathfrak{F}_\mu V^d \mathfrak{F}_\mu^\dagger)} + \frac{1}{\inf \text{spec}(\mathfrak{F}_\mu (V^d + \kappa V^{\text{od}}) \mathfrak{F}_\mu^\dagger)}. \end{aligned}$$

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Standard perturbation theory in κ .

$\leadsto$ conditions for “linear” versus “quadratic”.